

# TITLE PAGE

- ❖ **Problem Statement ID** – **SIH26084**
- ❖ **Problem Statement Title** – Real-Time Convective-Scale Nowcasting (0–6 h) of Severe Storms using Multi-Source Data Fusion
- ❖ **Theme** – Disaster Management / Smart Automation
- ❖ **PS Category** – Software
- ❖ **Team ID** – **85508 (SIH26084\_BUMBLEBEE)**
- ❖ **Team Name** – **Bumblebee**
- ❖ **Ministry / Agency** – Ministry of Earth Sciences (MoES) & India Meteorological Department (IMD)



GIST : BUMBLEBLE fuses radar, INSAT-3DR & lightning data on a 2 km grid to nowcast severe storms 0–6 h ahead, with live arrival countdowns.

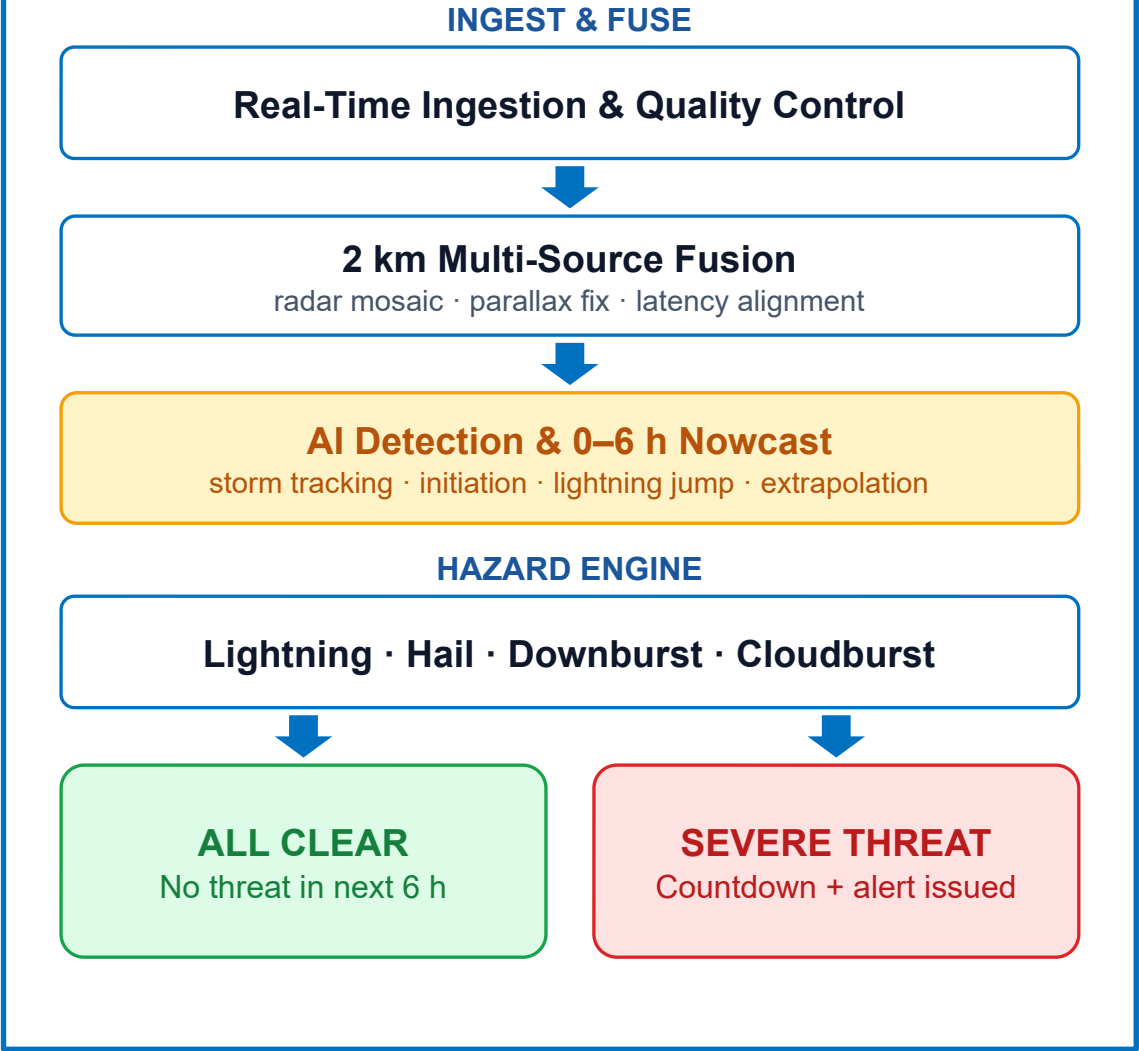
IDEA / SOLUTION

- ❖ **Multi-Source Data Fusion** – DWR radar, INSAT-3DR IR/WV & lightning on one 2 km grid
- ❖ **Early Storm Detection** – cloud-top cooling flags new storms before radar sees them
- ❖ **0–6 h Hazard Nowcast** – lightning, hail, gust & cloudburst every 10 min
- ❖ **Live GIS Dashboard** – arrival countdowns for 80+ cities, airports & farms

PROBLEM RESOLUTION

- ❖ **Lightning: India's #1 natural-disaster killer** → minute-level shelter alerts
- ❖ **Storms form in minutes** – too fast for coarse NWP; we nowcast at 2 km
- ❖ **Radar gaps** are filled by a satellite + lightning proxy
- ❖ **Vague district warnings** replaced by minute-level ETAs per place

BUMBLEBLE NOWCAST FLOW

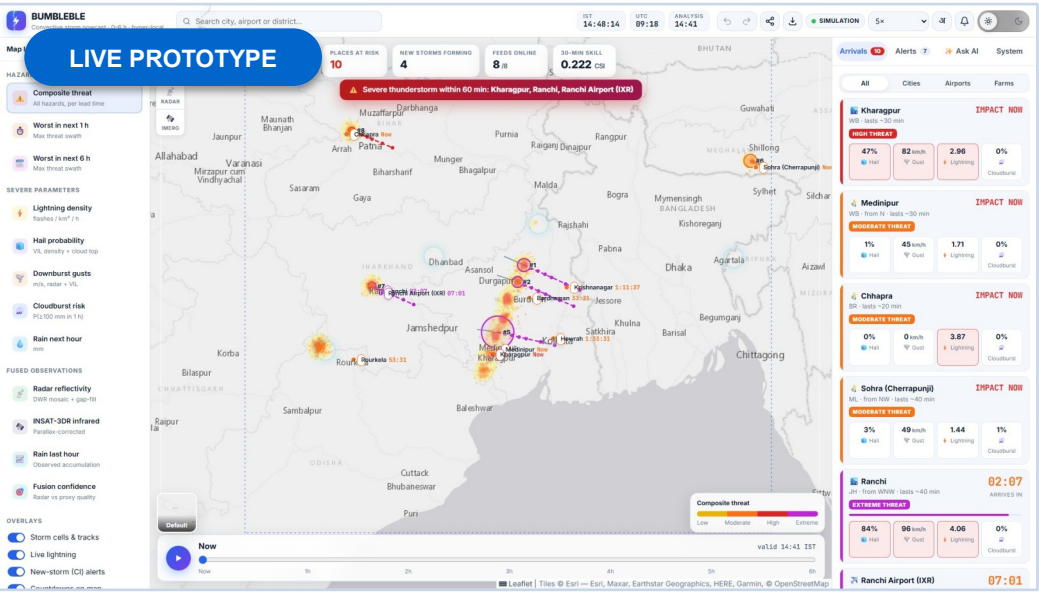


Live Demo

<http://localhost:8000> · [one-click Docker cloud deploy](#)

UNIQUENESS

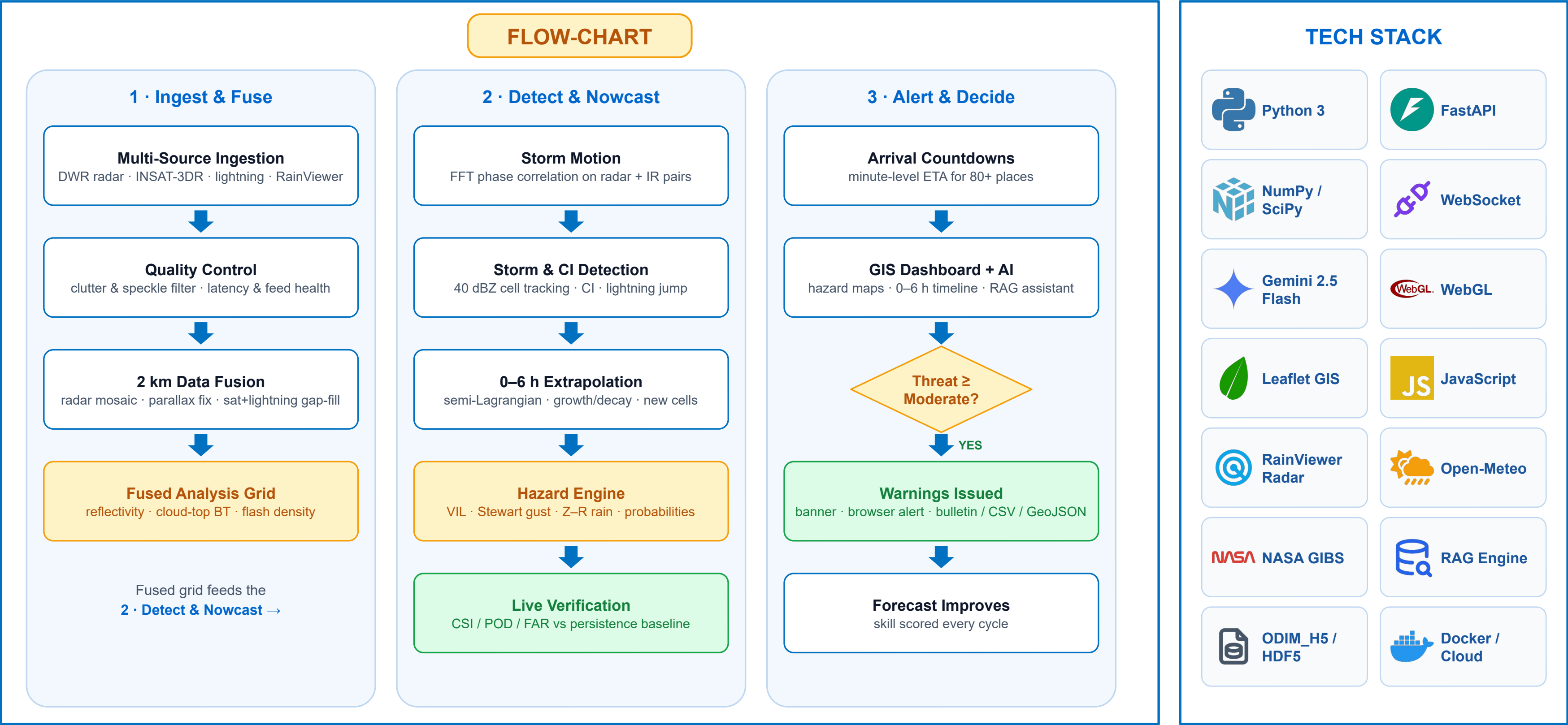
- ❖ **Minute-level countdowns** for 80+ places
- ❖ **Outage-proof:** satellite + lightning fill radar gaps
- ❖ **Self-verifying:** skill scored live vs baseline
- ❖ **AI + RAG assistant** (Gemini) in EN / हिन्दी



Live dashboard · simulated Nor'wester storm

Prototype Status: Working

9/9 automated tests · 150-min storm benchmark  
Realtime (live radar) + Simulated Nor'wester demo



## ANALYSIS OF FEASIBILITY

### Economic

- ❖ **Near-zero data cost:** free RainViewer, Open-Meteo & NASA feeds today
- ❖ **Funding:** MoES / IMD, NDMA & state disaster-tech programmes
- ❖ **Low infrastructure:** one CPU container — no GPU or supercomputer
- ❖ **Avoided losses** → lives, crops, flights & power lines protected

### Technical

- ❖ **Working prototype:** 9/9 tests pass; runs on live radar
- ❖ **<15 s per cycle** for all 37 lead times on a laptop CPU
- ❖ **Containerized** (Docker) → free cloud deploy (HF Spaces)
- ❖ **IMD-ready:** ODIM\_H5 radar, INSAT HDF5 & lightning push API

### Scalability

- ❖ **Region presets** for East, NE, North, Central, West & South India
- ❖ **Modular pipeline** → add radars, satellites & NWP blending
- ❖ **Open standards:** ODIM, GeoJSON, REST & WebSocket APIs

| Challenges & Risks                                                                                                       | How We Overcome Them                                |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|  <b>No INSAT / lightning feed yet</b> | Adapters built; MOSDAC & IMD/IITM data access       |
|  <b>Radar gaps &amp; outages</b>      | Satellite + lightning proxy with confidence score   |
|  <b>Skill drops after 2–3 h</b>       | Widening probabilities; NWP blending on roadmap     |
|  <b>AI overload &amp; API keys</b>    | Gemini fallback chain → demo mode; keys server-side |
|  <b>False alarms &amp; trust</b>      | Live verification; IMD warnings take precedence     |

## KEY METRICS

\* 150-min simulated Nor'wester benchmark, 24 verification samples

0.55

forecast skill (CSI)  
at 30 min · 7× baseline\*

86%

storm areas caught  
at 30 min (POD)\*

15 min

new storms flagged  
before radar\*

<15 s

full 0–6 h forecast  
on a laptop CPU

# IMPACT AND BENEFITS



## Social

Alerts in English & हिन्दी for all



## Safety

Early shelter from lightning & squalls



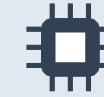
## Environmental

Climate-resilient cloudburst alerts



## Economical

Protects crops, flights & power



## Technological

AI data fusion + live GIS nowcasts

### WHO BENEFITS

#### District & State Disaster Authorities

- ❖ **Targeted Warnings** – **ETA per city & block**, not whole districts
- ❖ **Faster Response** – teams deployed before the storm arrives

#### Airports & Aviation

- ❖ **Gust & Wind-Shear Alerts** – live countdown for each airport
- ❖ **Safer Operations** – plan holds & diversions in advance

#### Farmers & Rural Communities

- ❖ **Hail & Cloudburst Warnings** – protect crops & livestock
- ❖ **Lightning Safety** – shelter advice in the local language

### NATIONAL IMPACT

#### Lives & Safety

- ❖ **Fewer Casualties** – lightning & storms kill thousands yearly
- ❖ **Disaster Resilience** – strengthens IMD & NDMA early warnings

#### Economic Growth

- ❖ **Crop Protection** – timely hail & heavy-rain alerts cut losses
- ❖ **Aviation & Power** – fewer diversions and line outages

#### Digital Leadership

- ❖ **Mission Mausam** – supports India's weather-ready goals
- ❖ **Open Integration** – GeoJSON, ODIM & REST plug into existing systems

## RESEARCH BASE

### Nowcasting & Storm Tracking

- ❖ **Radar Extrapolation** – Germann & Zawadzki (2002), Mon. Wea. Rev.
- ❖ **pySTEPS Nowcasting** – Pulkkinen et al. (2019), Geosci. Model Dev.
- ❖ **Cell Tracking** – Dixon & Wiener (1993) TITAN, J. Atmos. Ocean. Tech.

### Convection & Hazards

- ❖ **Convective Initiation** – Mecikalski & Bedka (2006) SATCAST, MWR
- ❖ **Lightning Jump** – Schultz et al. (2009), J. Appl. Meteor. Climatol.

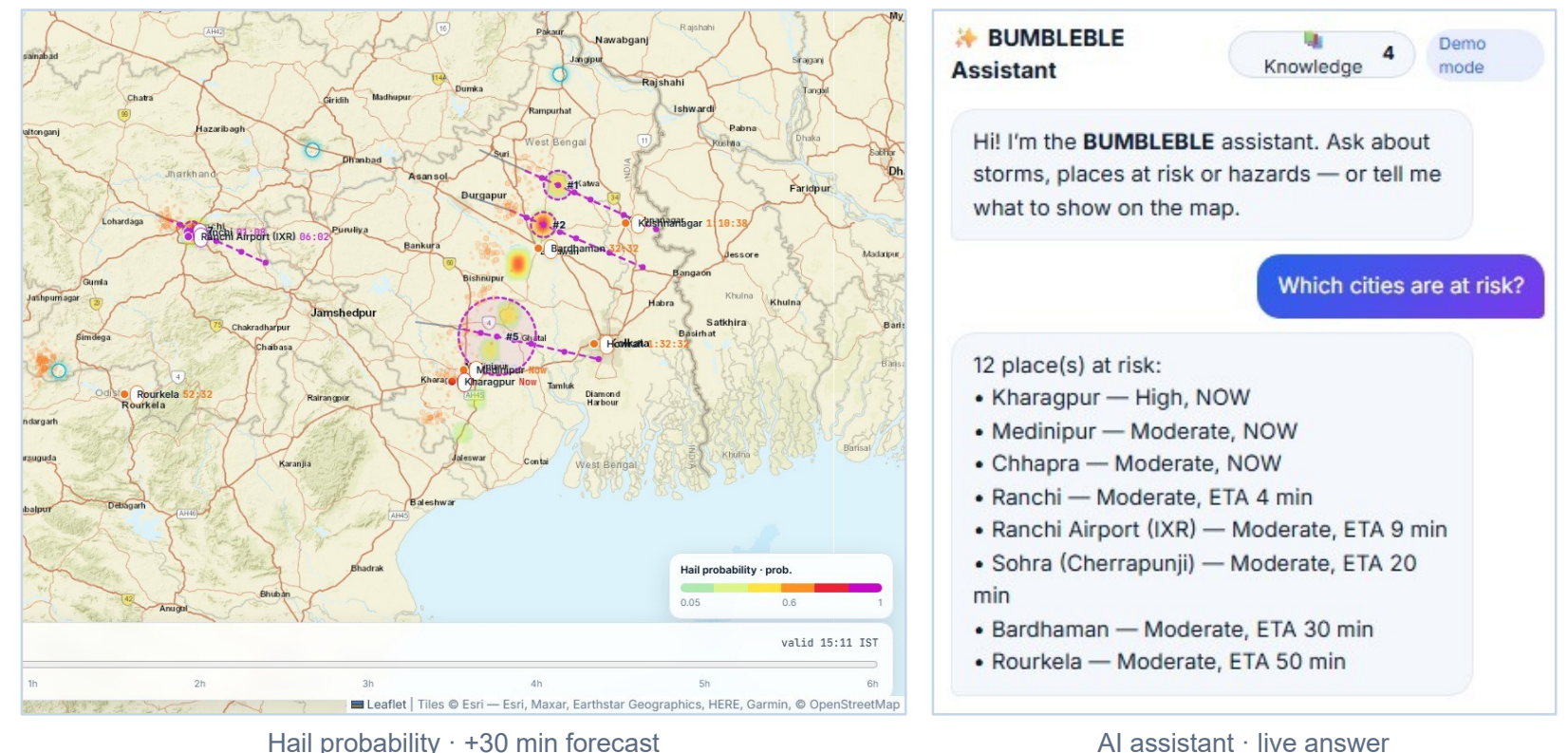
### Hazards & Data Standards

- ❖ **VIL Density Hail** – Amburn & Wolf (1997), Weather & Forecasting
- ❖ **Downburst Gust** – Stewart (1991) VIL + echo-top relation
- ❖ **Data Standards** – EUMETNET ODIM\_H5 radar; ISRO MOSDAC INSAT L1B
- ❖ **IMD Criteria** – rainfall classes, squall & cloudburst ( $\geq 100$  mm/h)

### Proven Implementations & Benchmarks

- ❖ **Benchmarked vs** – NOAA MRMS, UK Met Office STEPS, NCAR TITAN
- ❖ **Operational peers** – EUMETSAT NWC SAF CI, IMD district nowcasts

## WORKING PROTOTYPE · LIVE PROOF



**9/9**

automated tests pass

**37**

forecast steps, 0–6 h

**80+**

places with countdowns

- ❖ **Verified skill** – CSI 0.55 at 30 min, 7× persistence\*
- ❖ **Early warning** – new storms flagged ~15 min before radar\*
- ❖ **Real data** – live RainViewer radar frames every 10 min
- ❖ **Deploy anywhere** – one Docker container · free cloud hosting

\* 150-min simulated Nor'wester benchmark